SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

abhiyantriki 2026

HACKTOBERFEST 26

HACKDAY MUMBAI

X

ISTE KJSSE

EVENT TIMELINE

Time (IST)	Activity
8:30 AM	Reporting and Registration (closes at 9:30 AM)
10:00 AM	Opening Ceremony and ISTE introduction
10:30 AM	Problem Statements Revealed; Development Begins (First Half)
1:00 PM	Lunch break
1:30 PM	Development Resumes (Second Half)
4:30 PM	Development Ends; Final Submission Deadline
4:45 PM	Judging begins
6:45 PM	Judging Ends and Closing Ceremony

KEY REMINDERS

- Registration verification is mandatory.
- Problem statements will be announced at 10:30 AM.
- Hackathon development ends at 4:30 PM.
- Judging takes place from 4:45 PM to 6:45 PM.

What is Hacktoberfest?
Hacktoberfest is a global celebration of open source that takes place throughout October. It began in 2014 as a way to encourage people to contribute to open-source software and has grown into an international developer tradition. In 2026, Hacktoberfest is taking a new direction: its programme focuses on learning, experimenting, and building with open-source AI and open-weight models. Powered by Major League Hacking (MLH) and DEV, in partnership with DigitalOcean, the celebration includes online activities and local, in-person Fests around the world.

Hacktoberfest Comes to Mumbai
This edition is being presented as the first Hacktoberfest Hack Day hosted at Somaiya and a new opportunity for Mumbai's student developer community to take part in the wider Hacktoberfest movement. The event identifies K. J. Somaiya School of Engineering in Vidyavihar as the venue.

About ISTE KJSSE?
The Indian Society for Technical Education (ISTE) is a national, non-profit professional society focused on technical education, teacher development, student growth, and the development of engineering talent. ISTE KJSSE is the student chapter affiliated with K. J. Somaiya School of Engineering.

The 2026 Edition at a Glance

- Event: Hacktoberfest Hack Day Mumbai × ISTE KJSSE
- Date: 15 October 2026
- Venue: K. J. Somaiya School of Engineering, Somaiya Vidyavihar University, Vidyavihar East, Mumbai
- Format: An in-person Hack Day centred on building and demonstrating projects
- Published challenge themes: Best Use of Gemma 4 and Best Open-Source AI Project

Why this Collaboration Matters
Open source grows when people learn from one another and build things that

others can inspect, adapt, and improve. By bringing Hacktoberfest to Somaiya with ISTE KJSSE, the event links a global developer initiative with a local student community and gives participants a chance to move from learning about open-source technology to experimenting with it themselves.

Building Resilient Open-Source Software with AI

Background

Open-source projects often depend on third-party packages that can become outdated, unmaintained, deprecated, vulnerable, incompatible with newer versions, or difficult to replace. Maintainers may discover these problems only after something breaks, making dependency maintenance time-consuming and difficult to verify.
At the same time, open-source AI can assist developers with software maintenance tasks. However, AI-generated recommendations alone are not sufficient. A reliable developer tool should connect AI-assisted analysis with actual repository inspection, execution, and verifiable evidence.

The Challenge

Build an original, open-source AI-powered developer tool that helps maintainers identify, understand, or resolve dependency and maintenance problems in a real software repository. Your solution must work on the organizer-provided target repository and should perform its analysis dynamically rather than relying on pre-written answers or pre-generated outputs.

GITHUB REPOSITORY Link: https://github.com/rishrek/LinkWatch

(NOTE: THE REPO IS BUILT ON PYTHON 3.12 AND 3.13 ONLY. NOT COMPATIBLE WITH PYTHON 3.14)

Participants must choose one of the following two tracks.

Track 1 : Gemma 4 : Dependency Resilience

REQUIREMENT:

For teams competing specifically for Best Use of Gemma 4, Gemma 4 must be a meaningful part of the solution. Gemma 4 may be accessed through an appropriate supported method, including hosted API access.
Teams should identify the model, version, access method, and applicable terms in their documentation.

Build an AI-powered developer tool that helps a maintainer identify dependency risks and safely prepare for a dependency replacement. Help maintainers make dependency migration more transparent, testable, and reusable, reducing the risk of breaking otherwise healthy open-source software.

Your tool must analyse the target repository and:

1. Identify potentially at-risk dependencies using available repository and dependency metadata.
2. Select all identified dependencies and determine where and how it is used in the repository.
3. Propose a suitable replacement and migration approach for all dependencies identified.
4. Where technically feasible, execute tests against the original dependency and the proposed replacement.
5. Compare the observed results and identify compatibility gaps or unresolved risks.
6. Generate a reusable Dependency Exit Capsule containing the evidence and migration information.

The Dependency Exit Capsule should include, where applicable:

- Dependency details
- Risk indicators
- Usage locations
- Proposed replacement
- Migration steps
- Suggested tests
- Executed test results
- Compatibility gaps
- Open concerns and limitations

AI-generated recommendations must be clearly separated from information that was actually observed or verified through execution.

Track 2: Maintenance Agent Skill

REQUIREMENT:
The project must use open-source or open-weight AI as a meaningful part of the solution. Teams may choose an appropriate model or model family according to its applicable license and usage terms. Teams are not required to train, fine-tune, or create their own AI model.

If an Agent Skill is submitted, it should follow the applicable Agent Skills Open Standard requirements.
Follow: https://agentskills.io/home

Build an original, reusable AI-powered maintenance capability that performs a specific software-maintenance task on the organizer-provided target repository.

The capability may be implemented as:

- An Agent Skill
- A custom AI tool
- An automated maintenance workflow
- A meaningful extension to an existing open-source AI agent or model harness

The capability must analyse or interact with the target repository and:

1. Identify and understand a specific software-maintenance problem in the repository.
2. Perform a clearly defined maintenance task using the repository's actual code, dependencies, configuration, tests, or other relevant project data.
3. Use AI meaningfully to analyse the problem, provide recommendations, or assist in carrying out the maintenance task.
4. Where technically feasible, execute relevant tests, checks, or other verification steps to validate the result.
5. Identify errors, compatibility gaps, unresolved risks, or limitations based on the observed results.
6. Clearly distinguish AI-generated recommendations from information actually observed or verified through repository inspection or execution.
7. Generate a reusable Dependency Exit Capsule containing the findings, actions, evidence, and maintenance information produced during the task.

The Dependency Exit Capsule should contain, where applicable:

- Maintenance problem identified
- Relevant dependency or repository details

- Risk indicators
- Usage locations
- AI-generated analysis or recommendations
- Proposed changes or maintenance approach
- Suggested tests or verification steps
- Executed test results
- Compatibility or maintenance gaps
- Unresolved concerns and limitations

ORIGINAL CONTRIBUTION

If an existing framework or AI agent such as **Hermes Agent** is used, the team's contribution. Simply deploying an existing agent, changing its prompt, or modifying its user interface is not sufficient.

The submitted capability must add a new skill, tool, workflow, functionality, or meaningful modification that addresses the chosen maintenance problem.

(NOTE: There are many security vulnerabilities in the Github Repository. THOSE CONTAIN NO POINTS AND NEED NOT BE CONSIDERED AS OF NOW.)

PRE-EVENT REQUIREMENTS

Participants are requested to complete the following preparations before the hackathon to ensure a smooth and productive development experience.

1. Bring your laptop, charger, and any required adapters or accessories.
2. Create or verify access to your GitHub account. Participants must be able to create public repositories, push code, and collaborate with their teammates.

   FINAL GITHUB REPO Name: H26_TeamName_ProjectName

3. Development Environment: Install and test the required code editor, programming languages, package managers, Git, and other tools relevant to your planned project.
4. AI Model and API Access: Identify your intended AI model and verify access before the event.
   - For Track 1, participants must use the Gemma 4 open-weight model.
   - For Track 2, participants may choose a suitable model and agent framework. Test any required API access, credentials, local model setup, and usage limits in advance.

5. Read the selected track's problem statement, expected deliverables, and evaluation criteria before the event.
6. Complete event registration and confirm your team details according to the event's team formation rules.
7. Open-Source Readiness: Familiarize yourself with basic Git and GitHub workflows, including commits, branches, and repository publishing. Be prepared to publish your final project under an appropriate open-source license.
8. Optional Framework Setup : Participants planning to use tools such as Hermes Agent or other AI frameworks may explore their documentation and setup instructions beforehand. These tools are optional, not mandatory.

Important Notes

- Do not commit API keys, passwords, private tokens, or other sensitive credentials to public repositories.
- Check model availability, API quotas, hardware requirements, and applicable model usage terms in advance.

- Teams should develop their own original solution during the hackathon. Existing libraries and open-source frameworks may be used in accordance with their licenses.
- Participants should arrive prepared to build, test, and demonstrate a functional prototype within the six-hour duration.
- Cloud services, paid APIs usage is strictly not allowed.

For more information on usage of Gemma4 and DigitalOcean, refer to subparts of this tab.

About Gemma 4 - TRACK 1

Gemma is a collection of open-weight, high-performance foundation models developed by Google DeepMind and Google's AI teams. The Gemma family is built upon the same research, training data, and technical architectures used for Google's flagship proprietary model series, Gemini.

Gemma 4 represents the generational leap in Google DeepMind's open-weights portfolio. While earlier iterations focused primarily on text generation, Gemma 4 is built from the ground up as a natively multimodal, reasoning-centric foundation family released under an open Apache 2.0 license.

Unlike fully proprietary closed-API models, Gemma provides open weights under commercially permissible terms, enabling researchers, engineers, and organizations to download, inspect, fine-tune, and deploy state-of-the-art models locally on edge hardware or within private cloud environments.

Click here to View Track 1

HOW TO USE GEMMA 4 API KEYS

1. Take up the Gemma Quickstart on the MLH.
   Click Here

2. Start your journey today by Gemma API Docs
   Click Here

3. Here is your Beginner Guide
   Click Here

About Digital Ocean : TRACK -2

About DigitalOcean

DigitalOcean is a cloud computing platform that provides developers and organizations with tools to build, deploy, host, and manage applications over the internet.
DigitalOcean serves as a reliable deployment platform for turning ideas into working prototypes. It can be used to host websites, backend APIs, databases, and AI-powered applications, making their projects accessible through a public URL for users.

About Hermes

Hermes Agent is an open-source AI agent developed by Nous Research. It is designed to go beyond answering questions and actually help users perform tasks. It can use tools, execute code, interact with external services, and automate multi-step workflows. Unlike a basic chatbot, Hermes Agent is designed to work in a persistent environment, where it can maintain useful context, learn from experience, and build reusable skills.

Click Here to View Track 2

HOW TO USE DigitalOcean AS PART OF YOUR TECH STACK:

1. Try DigitalOcean's AI tools
   Click Here

2. Browse DigitalOcean libraries
   Click Here

3. Deploy Hermes in one click
   Click Here

NOTE: If this existing framework i.e, Hermes Agent is used, the team's contribution must be meaningful and original.
Simply deploying an existing agent, changing its prompt, or modifying its interface is not sufficient.

EVALUATION METRICS

Participants Will Be Evaluated On The Following Criteria:

1. Problem Relevance and Usefulness
2. Technical Implementation and Functionality (identification of dependencies)
3. Meaningful AI Integration
4. Proof, Testing, and Reliability
5. Originality of Contribution
6. Open-Source Quality and Documentation

Additional information:

Projects will be assessed on:

- Problem relevance: Importance of the developer or maintainer problem being addressed.
- Technical implementation: Functionality, feasibility, and quality of the prototype.
- Meaningful AI integration: Appropriate use of AI to support the solution.
- Verification and reliability: Evidence supporting outputs and honest treatment of limitations.
- Open-source value: Reusability, documentation, and potential for community adoption.
- Originality: A meaningful contribution beyond existing tools or default framework functionality.

Rules and Regulations

1. Eligibility & Teams

- Teams must have 3 to 4 members.
- Each participant may be part of only one team.
- Every participant must bring their own ID card.
- Every team member must contribute meaningfully.

2. Building Rules

- Problem statements will be revealed at the official start of the hackathon.
- Development time is 10:30 AM to 4:30 PM.
- Teams must use only Git and maintain a meaningful commit history.
- Teams must ensure their project works and can be demonstrated.

3. Pre-existing Work

The core project must be developed during the official hackathon period.

- Pre-existing code, features and assets must be disclosed appropriately.
- Third-party libraries, models, datasets and APIs must be properly acknowledged.
- Misrepresenting pre-existing or third-party work is prohibited.

4. AI & Gemma 4

AI tools are allowed and may be used for development, learning, debugging, documentation and other legitimate purposes.

- AI-generated outputs must be verified, and heavy use of AI must be avoided.

Best Use of Gemma 4

- Projects competing for Best Use of Gemma 4 must integrate Gemma 4 meaningfully.

Integration of Digital Ocean

- Projects must use Hermes agent for their AI Agents for competing the Maintenance Agent.

For more information, Refer to the Resources Tab.

5. Submission Requirements

- To qualify for judging, submit a correctly named, public GitHub repository forked from the official starter link. Your repository must include an open-source license and a complete README detailing the project description, tech stack, AI usage information, and local setup instructions.

6. Fair Play

Participants must not:

- Access another team's private code or credentials.
- Copy another team's project or deliberately damage another team's work.
- Manipulate judging.
- Falsify development history.
- Misrepresenting pre-existing or third-party work.
- Attack event infrastructure without explicit authorization.

7. Code of Conduct & Venue Rules

- Harassment, discrimination, threats and disruptive behaviour are prohibited.
- Participants must follow venue instructions and respect campus property.
- Weapons, illegal drugs and dangerous items are prohibited.
- Participants are responsible for their personal belongings.
- Participants must respect privacy when photographing, filming or publishing content.

- Participants must not publish private or sensitive material of other attendees without appropriate permission.

8. Enforcement

- Organizers have the authority to enforce rules and resolve situations not explicitly covered.
- Violations may result in warnings, disqualification or removal from the venue.

Final Project Presentation & PPT Requirements

Make the github Repo name as : H26_TeamName_ProjectName

1. PPT Presentation format

PPT template link

NOTE: Please make a copy to edit. Do not modify the overall layout or visual theme.

Slide-wise Guidelines for Participants

Slide 1: Team Information & Track

- Team name and team members.
- Selected track and problem statement.
- Brief overview of the problem being addressed.

Slide 2: Solution & Workflow

- Proposed solution and workflow diagram.
- Key technologies, AI integration, and data sources.
- Verification process and expected outputs.

Slide 3: Technical Approach

- Technologies, tools, frameworks, and AI models.
- Methodology, algorithms, and system architecture.
- Implementation, testing, and validation.

Slide 4: Feasibility & Viability

- Technical feasibility and required resources.
- Potential challenges, limitations, and risks.
- Scalability, sustainability, and long-term viability.

Slide 5: Impact & Benefits

- Key benefits and target users.

- Social, economic, or environmental impact.
- Measurable outcomes and long-term benefits.

Slide 6: Research & References

- Relevant research papers and existing solutions.
- Credible sources and official documentation.
- Proper citations and working reference links.

2. Track-Specific Proof Requirements

Track 1

- Present relevant analysis and supporting evidence.
- Demonstrate the core functionality and its output.
- Show testing results and generated deliverables.
- Distinguish AI-generated suggestions from verified results.

Track 2

- Demonstrate the practical use of your AI capability.
- Showcase your original implementation.
- Provide execution evidence and error handling.
- Explain how others can reuse your contribution.

3. Live Prototype Demonstration

- The submitted project must include an appropriate open-source license, such as: MIT, Apache-2.0, etc. Teams must also comply with the licenses of all third-party libraries, frameworks, models, datasets, and tools used.
- Solution must accept and analyse the organizer-provided target repository. Teams must not rely exclusively on manually prepared sample inputs.
- README.md containing:
    1. Problem and solution overview

2. Setup instructions
3. Usage instructions
4. Architecture
5. Dependencies
6. AI model details [Model name,Model version, Model source, Access method, Applicable model license or usage terms, Role of the model in the project]
7. Known limitations

- Demonstrate a real input, core functionality, and actual output.
- Explain AI integration and your original contribution.
- Present evidence supporting the results.
- Disclose incomplete features; do not present mock-ups as working prototypes.
- Keep source code accessible to judges and use a short screen recording as a fallback for technical failures.